10.5 Matching Data Displays

Lesson Introduction

Benchmark	MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.	Learning Target	• I can match displays of data to the appropriate data set.

	Building On	Working Toward
Benchmark Coherence	MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.	MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

Essential Questions	• What is the difference between a box plot and histogram? • How can you correctly match a data set to a box plot? • How can you correctly match a data set to a histogram? • Which types of data are better to use with box plots and histograms?
Lesson Narrative	In this lesson, students begin by finding the five-number summary of a data set, then choose the box plot and histogram that displays the data correctly. Students explain which features of each display they utilized to find the match and then compare the box plot and histogram to decide which displays the data better. Students then match histograms to data sets and box plots before wrapping up with a growth mindset reflection.

Component Summary	10.5.1 Warm-Up (~5 min.)	Find the five-number summary and range for a set of data *(SE p. 222)*	10.5.3 Activity (10-15 min.)	Sort cards to match data set with histogram and box plot *(SE p. 224)*
	10.5.2 Exploration (10-15 min.)	Find the five-number summary and IQR with a partner to create box plot and histogram *(SE p. 222)*	10.5.4 Your Turn! (5 min.)	Find the five-number summary with IQR and match the data points to the correct box plot and histogram *(SE p. 224)*
	10.5.5 Wrap-Up (~5 min.)	Students reflect on their growth mindset throughout the lesson *(SE p. 225)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Box plot • Histogram	• Card sort • (Optional) Additional paper • (Optional) Computer	• The card sort will need to either be printed and cut or digitally altered to be given to each student. If this is not possible, project the cards in a way for the whole class to see them. • Consider creating partner-pairs before the lesson begins. • Before the lesson is presented to students, decide how and in what format the card sort will be presented.

Teacher Tips

Common Misconceptions

- Students may incorrectly match the data to a display by misreading the data.
- Students may calculate or interpret the IQR incorrectly.
- Students may not understand that certain information cannot be determined from a box plot/histogram.

Things to Consider

- Consider encouraging students to create their own box plots and histograms based on data sets before choosing their answers if they get stuck.
- Consider reviewing the features of the five-number summary to activate prior knowledge before students begin the lesson.

Literacy and Language Routines

Notice and Wonder

10.5.3 Activity provides students with the opportunity to make connections between data sets, histograms, and box plots. Engage students in a Notice and Wonder routine before they begin matching to encourage students to draw conclusions about the data and consider what the shape, center, and spread should look like. This will help students accurately match the visual displays and data sets.

Mathematical Thinking and Reasoning (MTR) Standard

MA.K12.MTR.2.1: Multiple Representations

Students analyze and present data in a variety of ways throughout this lesson. Clarifications for this standard includes showing students that various representations can have different purposes and can be useful in different situations. This is particularly evident as students consider what information can be drawn from box plots versus histograms, understanding that both representations may be better suited for different purposes.

Differentiate Instruction

Displaying the data on histograms and box plots provides multiple visual entry points for students, especially those who may struggle to visualize and interpret the data on their own. Continue to encourage students to represent the data visually as they work through this lesson, particularly when trying to match data sets to box plots and histograms. Creating their own visual displays will deepen understanding and provide an additional way to verify responses.

Lesson Notes:

10.5.1 Warm-Up: Bell Work

Component Overview: *Students determine the five-number summary and range for a set of data in context.*

10.5.1 Warm-Up: Bell Work

1. During the 2020 NBA Season, Jimmy Butler played in the Championship Game against the Los Angeles Lakers. The total points scored by Jimmy Butler for the conference semifinals, conference finals, and championship games are listed below.

   $12, 35, 22, 40, 25, 23, 22, 17, 24, 17, 14, 20, 17, 17, 30, 13, 40$

   a. Determine the five-number summary for the data set.

      Min: 12

      Q1: 17

      Median: 22

      Q3: 27.5

      Max: 40

   b. Determine the range.

      28

Observe students as they work to determine if additional prompting or support may be needed. Some students may forget that it is helpful to first rewrite the data set in order from least to greatest when determining the five-number summary.

Consider asking the following questions, if needed:

- *What are the five measures included in the five-number summary?*
- *When determining the five-number summary and range, what is helpful to do with the values in the data set first?*

10.5.2 Exploration: Matching Displays

Component Overview: *In this Exploration, students will work with a partner to find the five-number summary and the interquartile range for the given set of numbers. With their partner, students will determine which displays correctly represent the data set and discuss which display is best to represent the given data.*

10.5.2 Exploration: Matching Displays

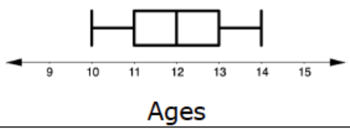
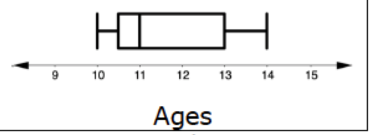
Work with your partner to complete the following:

A middle school band director collects the ages of each clarinet player in the band, shown below.

$$10, 10, 11, 11, 11, 12, 12, 12, 13, 13, 14$$

1. Determine the five-number summary and IQR for the ages of the clarinet players in the band.
   Minimum: 10, *Maximum:* 14, *Quartile 1:* 11, *Median:* 12
   Quartile 3: 13, *IQR:* 2

2. Which box plot accurately displays the ages of the clarinet players?
   Box Plot A

Box plot A	Box plot B
	

3. Discuss your choice with your partner and summarize.
   Answers will vary. The correct box plot can be quickly identified because the median is incorrect in Box plot B.

In a box plot, whiskers extend from each quartile to the minimum or maximum. The left whisker represents the bottom 25% of the data and the right whisker represents the top 25%.

4. Discuss with your partner how the location of the median and the length of the whiskers can be used to determine the shape of the distribution. Summarize your discussion.
   Answers will vary. If the length of each whisker is about the same, the shape is roughly symmetric. If one whisker is longer than the other, the shape is skewed.

MA.K12.MTR.2.1: Multiple Representations
In this activity, students will work with a partner to determine appropriate representations of the given data in different ways, including: a box plot, a table, and a histogram.

For students who are struggling to explain how the median and whiskers can be used to determine the shape of the distribution, ask questions like:

- *What does the median tell you about the data set? How does this impact the shape of the distribution?*
- *What do the length of the whiskers tell you about the data set? How does this impact the shape of the distribution?*

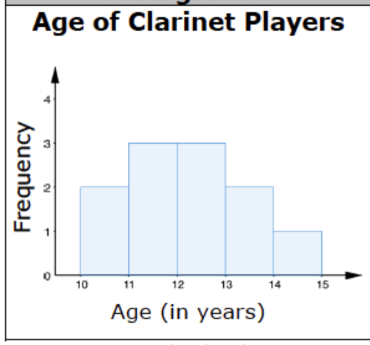
10.5.2 Exploration: Matching Displays (continued)

5. Create a frequency table of the data set.

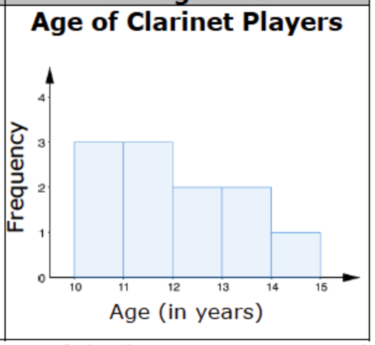
Age Range	Number of Clarinet Players
$10 - 11$	2
$11 - 12$	3
$12 - 13$	3
$13 - 14$	2
$14 - 15$	1

6. Which histogram most accurately displays the data?
Histogram A

Histogram A



Histogram B



7. Discuss which characteristics of the histogram you used to make your selection.
Frequency data.
8. Describe the shape of the histogram you chose, including any gaps, outliers, or clusters.
Answers will vary. Histogram A has a bell-shaped curve. The data is evenly disbursed. It is clustered about ages $11 - 13$ with no real outliers.
9. Compare the box plot selected for the data with the histogram selected. Discuss which display better represents the data. Summarize your discussion.
Answers will vary. The box plot represents the data better because you can see the ages are clustered around $11 - 13$. The histogram represents the data better because you can see the frequency of the ages of student clarinet players.

This activity provides multiple entry points through visual displays of data and partner discussion.

Challenge students to come up with examples of types of data that would be better displayed in a box plot or histogram and explain why.

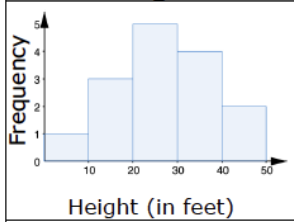
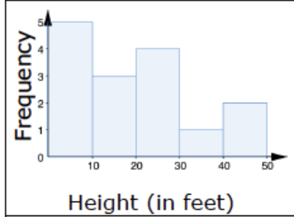
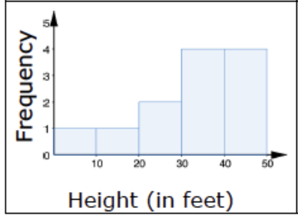
10.5.3 Activity: Card Sort

Component Overview: *In this activity, students work collaboratively to match histograms with the raw data sets used to create them and box plots that represent the same data through a card sort. If printing the cards is not an option, project the cards for the whole class to see.*

10.5.3 Activity: Card Sort

1. An arborist visited a pine tree farm and took measurements of the tree heights, in feet, in three different fields. He dropped his paperwork and all of the data he collected got mixed up.

   Complete the table to match each histogram with its corresponding data set and box plot. Write the number and letter from the card that corresponds to each match.

Histogram	Data Set	Box Plot
	*1*	*H*
	*2*	*B*
	*3*	*Y*

2. Compare your answers with your partner. Make any revisions, if needed.

Notice and Wonder
Before the card sort, consider engaging students in a Notice and Wonder routine. Encourage students to draw conclusions about the data by asking questions like:

- *What features did you use to match the displays and data?*
- *Did you start by matching data to a display or matching the displays to each other? Why?*

The card sort will help provide kinesthetic learners with an entry point into the lesson. If students are matching incorrectly, consider prompting to justify their choices.

10.5.4 Your Turn!

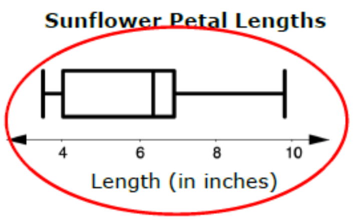
Component Overview: *In 10.5.4 Your Turn!, students work independently to match a box plot and histogram to a given set of data.*

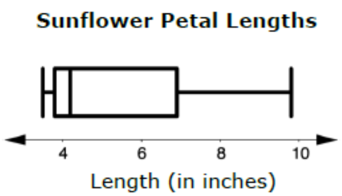
10.5.4 Your Turn!

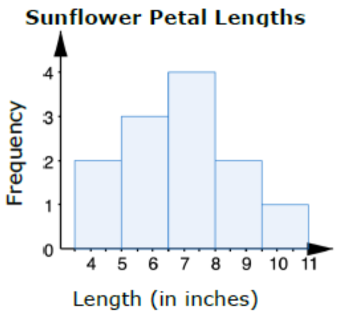
1. A horticulturist is researching sunflowers to find out the lengths of the petals on a sunflower. She gathers information from 12 sunflowers and measures the length of the petals, in inches, on each flower.

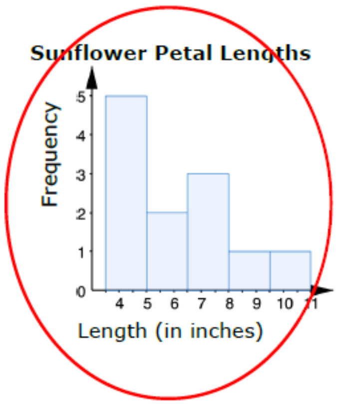
   6.8, 6.4, 6.2, 3.5, 3.6, 4.1, 3.9, 4.5, 6.7, 7.2, 8.1, 9.8

   Circle the boxplot and the histogram that best display the data set.









To ensure students are thinking through each choice and not simply guessing, ask questions like:

- *What is the median of the data set? How do you know? (Finding the median will allow students to identify the correct box plot.)*
- *How would you describe the spread of this data? Is it symmetrical? Why or why not? (When writing the data in order, students may observe that there are many small values in comparison to larger values to help identify the correct histogram.*

For students who finish early, ask them to explain which display better represents the data and why. This could also include asking which features of the data are made most clear in each representation.

10.5.5 Wrap-Up: Growth Mindset

Component Overview: *Students will reflect on their growth mindset throughout the lesson. Encourage students to answer with fidelity to find areas of weakness that can be targeted in a later lesson.*

10.5.5 Wrap-Up: Growth Mindset

1. Complete the table as you reflect on today's lesson.
   Answers will vary.

Which mindset did I demonstrate?	Check the best description.
Did I use the partner discussions as learning opportunities?	o Never o Sometimes o Usually o Always
Did I work on problems that challenged me?	o Never o Sometimes o Usually o Always
Did I use strategies to "un-stick" myself when I found the problems difficult?	o Never o Sometimes o Usually o Always

To provide additional means of observation and reflection, encourage students to find and name examples for the mindsets they demonstrated. For example, pointing out a strategy they used in the lesson to "unstick" themselves.